

Topic 02: Slope Fields, Solution Curves & Qualitative Behavior

Professor Leonard Master Series · Lessons 5 & 6

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: Seeing Without Solving

Professor Leonard begins qualitative analysis with an eye-opening question:

"What if an equation is so nasty that no algebraic trick on Earth can integrate it? Do we give up? Absolutely not! The differential equation $dy/dx = f(x, y)$ is an explicit formula for slope. At every single coordinate (x, y) in the Cartesian plane, the equation hands you a tiny tangent vector. If you drop a particle into that ocean of vectors, the fluid current carries it along the exact solution curve!"

A **slope field** (or direction field) is a visual grid of short line segments whose slopes equal $f(x, y)$.

- **Isoclines:** Curves along which the slope is constant: $f(x, y) = c$.
 - **Equilibrium Solutions:** Horizontal lines $y = c$ where $f(x, y) = 0$ for all x . Solutions starting on an equilibrium line stay there forever.
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2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 2.1: Isocline Construction & The Straight-Line Solution

Statement:

1. Sketch the slope field for $\frac{dy}{dx} = x - y$ using the method of isoclines for $c = -1, 0, 1, 2$.
2. Determine if any solution curve is a straight line $y = mx + b$.

Step-by-Step Whiteboard Solution:

- **Step 1: Set up the isocline equations $x - y = c$:**

$$y = x - c$$

- For $c = 0$: $y = x$. Along the line $y = x$, all slope segments are perfectly horizontal ($m = 0$).
- For $c = 1$: $y = x - 1$. Along this line, every tangent segment has slope $m = 1$.
- For $c = -1$: $y = x + 1$. Tangent slope is $m = -1$.
- For $c = 2$: $y = x - 2$. Tangent slope is $m = 2$.

- **Step 2: Check for a straight-line solution** $y = mx + b$: If $y = mx + b$, then $dy/dx = m$. Substitute into the ODE:

$$m = x - (mx + b) \implies m = (1 - m)x - b$$

This must hold for all x . Equating coefficients:

- $1 - m = 0 \implies m = 1$.
- $-b = m \implies b = -1$.

- **Step 3: Professor Leonard's Epiphany:**

$$y(x) = x - 1$$

Notice that the isocline for $c = 1$ is

identically

the line $y = x - 1$. Because every slope tick mark on this line has slope 1, the curve never leaves the line! It is an exact straight-line solution!

- **Step 4: Asymptotic Funnel:** Any solution starting above $y = x - 1$ bends down toward it; any solution starting below bends up toward it. As $x \rightarrow \infty$, all solution curves converge asymptotically to $y = x - 1$.

Problem 2.2: Autonomous ODE & Stability Classification

Statement: For the autonomous differential equation $\frac{dy}{dx} = y(y - 2)(y + 1)$:

1. Find all equilibrium solutions.
2. Construct the one-dimensional phase line.
3. Classify each equilibrium as asymptotically stable (sink), unstable (source), or semi-stable (node).
4. Sketch the qualitative behavior of solutions $y(x)$ for various initial values.

Step-by-Step Whiteboard Solution:

- **Step 1: Find critical equilibrium points:** Set $\frac{dy}{dx} = 0 \implies y(y - 2)(y + 1) = 0$. Equilibrium solutions are the three horizontal lines:

$$y = -1, \quad y = 0, \quad y = 2$$

- **Step 2: Sign analysis across the intervals:**

- **Interval** $(2, \infty)$: Test $y = 3 \implies (3)(1)(4) = +12 > 0$. Slopes are positive ($\uparrow$). Solutions grow to $+\infty$.
- **Interval** $(0, 2)$: Test $y = 1 \implies (1)(-1)(2) = -2 < 0$. Slopes are negative ($\downarrow$). Solutions fall toward $y = 0$.
- **Interval** $(-1, 0)$: Test $y = -0.5 \implies (-0.5)(-2.5)(0.5) = +0.625 > 0$. Slopes are positive ($\uparrow$). Solutions rise toward $y = 0$.
- **Interval** $(-\infty, -1)$: Test $y = -2 \implies (-2)(-4)(-1) = -8 < 0$. Slopes are negative ($\downarrow$). Solutions dive toward $-\infty$.

- **Step 3: Classify stability on the Phase Line:**

- At $y = 2$: Arrows point away ($\uparrow$ above, $\downarrow$ below) $\implies$ **Unstable (Source)**.

- At $y = 0$: Arrows point toward $y = 0$ ($\downarrow$ from above, $\uparrow$ from below) $\implies$ **Asymptotically Stable (Sink / Attractor)**.
- At $y = -1$: Arrows point away ($\uparrow$ above, $\downarrow$ below) $\implies$ **Unstable (Source)**.

Problem 2.3: Second Derivative Test & Inflection Points of Solution Curves

Statement: For the autonomous ODE $\frac{dy}{dt} = y^2 - 4$:

1. Determine the concavity of the solution curves in terms of y .
2. Find any inflection points where the concavity changes.

Step-by-Step Whiteboard Solution:

- **Step 1: Differentiate both sides with respect to t using the Chain Rule:**

$$\frac{d^2y}{dt^2} = \frac{d}{dt} (y^2 - 4) = 2y \frac{dy}{dt}$$

- **Step 2: Substitute $dy/dt = y^2 - 4$ into the second derivative:**

$$\frac{d^2y}{dt^2} = 2y(y^2 - 4) = 2y(y - 2)(y + 2)$$

- **Step 3: Analyze concavity across regions:**

- $y > 2$: $2y(y - 2)(y + 2) > 0 \implies$ **Concave Up**.
- $0 < y < 2$: $y > 0, y - 2 < 0, y + 2 > 0 \implies \frac{d^2y}{dt^2} < 0 \implies$ **Concave Down**.
- $-2 < y < 0$: $y < 0, y - 2 < 0, y + 2 > 0 \implies \frac{d^2y}{dt^2} > 0 \implies$ **Concave Up**.
- $y < -2$: all three factors negative $\implies \frac{d^2y}{dt^2} < 0 \implies$ **Concave Down**.

- **Step 4: Inflection points:** Inflection occurs where $\frac{d^2y}{dt^2} = 0$ while $dy/dt \neq 0$. $2y = 0 \implies y = 0$. As curves cross $y = 0$, they change from concave down to concave up!

Problem 2.4: Circular Slope Field & Orthogonal Trajectories

Statement: Sketch and analyze the slope field for $\frac{dy}{dx} = -\frac{x}{y}$. Show that solution curves form concentric circles.

Step-by-Step Whiteboard Solution:

- **Step 1: Evaluate slopes along rays:**

- On the x -axis ($y = 0$): slope is undefined (vertical tangents).
- On the y -axis ($x = 0$): slope is 0 (horizontal tangents).
- Along the line $y = x$: slope is $-x/x = -1$.
- Along the line $y = -x$: slope is $-x/(-x) = +1$.

- **Step 2: Analytical verification:**

$$ydy = -xdx \implies \int ydy = -\int xdx \implies \frac{y^2}{2} = -\frac{x^2}{2} + C_1 \implies x^2 + y^2 = C$$

The solution curves are concentric circles of radius $R = \sqrt{C}$ centered at the origin.

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Forgetting the Chain Rule for Concavity:** Students write $\frac{d^2y}{dt^2} = 2y$, completely forgetting $\frac{dy}{dt}$! You must multiply by y' via the chain rule: $y'' = 2y \cdot y'$.
- **Trap 2: Semi-stable Equilibria:** If an equilibrium has the same sign on both sides (e.g., $y' = (y - 1)^2$), arrows point up on both sides. A solution from below hits $y = 1$, but a solution from above moves away. This is **semi-stable**.